

Vance Vanvolkenburgh

655.662 — Introduction to Healthcare Systems Engineering
Module 3 | M3A1: Op Needs for RCS

Operational Needs and Requirements — PD-RPM Remote Care System

An operational need states a deficiency from the clinician's or patient's point of view and names no solution; an operational requirement is the system's measurable answer to that need, written as a "shall" statement. The needs below are the gaps; each requirement traces to one of them and is stated so that it can be verified by test rather than by opinion.

Operational Needs

- ON-1** Need for the home dialysis nurse to detect a declining ultrafiltration trend across successive treatments without waiting for the next monthly clinic visit.
- ON-2** Need for the patient and care partner to complete nightly therapy with no additional operational burden beyond running the cyclor itself.
- ON-3** Need for the care team to recognize early peritonitis indicators — effluent cloudiness, abdominal pain, fever, exit-site change — within hours of onset rather than at the next scheduled contact.
- ON-4** Need for the nephrologist to evaluate whether a prescription change actually improved volume status, before the next scheduled visit rather than after it.
- ON-5** Need for the home program to anticipate a solution or consumable shortfall for a patient before that patient runs short.

Operational Requirements

- OR-1** The system shall compute per-session net ultrafiltration for each of the trailing 14 treatments and shall flag a declining trend when the 7-day mean falls more than 20 percent below the trailing 30-day mean. (*satisfies ON-1*)
- OR-2** The system shall transfer each completed treatment record automatically within 30 minutes of session end, requiring no patient interaction beyond normal cyclor operation. (*satisfies ON-2*)
- OR-3** The system shall present a structured daily symptom and exit-site check completable in under 60 seconds, and shall route any positive peritonitis screen to the on-call nurse within 15 minutes. (*satisfies ON-3*)

OR-4 The system shall associate each prescription change with the subsequent 14 days of ultrafiltration, weight, and blood-pressure data and present that comparison to the nephrologist in a single view. *(satisfies ON-4)*

OR-5 The system shall project days of supply remaining from observed consumption and shall notify the home program when projected supply for any patient falls below 14 days. *(satisfies ON-5)*

KPP Candidates

Alert Latency — A critical alert shall reach the monitoring nurse within 15 minutes of the triggering data.

Operational Availability (A_o) — The monitoring and alerting path shall maintain A_o of at least 0.98.

Data Completeness — At least 95 percent of completed cyclor sessions shall be ingested within 24 hours.

Patient Daily Interaction Time — Daily patient interaction shall not exceed 60 seconds.

Detection Lead Time — Median peritonitis flag shall precede patient-initiated contact by at least 24 hours.

False Alert Rate — Non-actionable alerts shall not exceed one per patient per month.

Six candidates are proposed, at the upper bound of the five-to-six rule of thumb. Alert Latency, Operational Availability, and Detection Lead Time measure whether the system delivers its clinical value; Patient Daily Interaction Time and False Alert Rate protect the two users whose tolerance determines whether the system is actually used. False Alert Rate is included deliberately: a monitoring system that meets every other parameter still fails if the nurse stops reading its alerts.